

Simulated Agency as Active Control:
A Unified Framework Integrating Choice Theory,
Perceptual Control Theory, the Free Energy Principle,
Sparse Semantic Projection, and Sheaf-Theoretic Coherence

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Abstract

This monograph develops a comprehensive theoretical synthesis unifying William Glassers internal control psychology, William Powers Perceptual Control Theory (PCT), Karl Fristons Free Energy Principle (FEP), and contemporary sparse-representation and sheaf-theoretic approaches to cognition. Although these traditions emerged independently, they converge on a single deep principle: agents maintain their integrity by minimising a discrepancy functional between an internal normative model and the sensory world. This can be achieved only through two operations: acting on the environment or updating the internal model. The dual-gradient structure of adaptive behaviour is shown to be mathematically inevitable given thermodynamic, energetic, and structural constraints. We formalise these correspondences through variational calculus, Riemannian geometry, sparse dictionary learning, and sheaf cohomology, demonstrating that PCT is a special case of active inference under linear Gaussian assumptions, that Glassers reorganisation corresponds to hyperprior updating, and that biological sparsity is a thermodynamically inevitable constraint induced by metabolic cost. The resulting framework provides a unified architecture for psychology, theoretical neuroscience, and the design of interpretable artificial agents.

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1 Introduction and Motivation

Across the twentieth century, several intellectual traditions independently converged on a strikingly similar picture of how agents, organisms, and cognitive systems operate. William Glasser's internal control psychology, originating in Reality Therapy and maturing into Choice Theory (Glasser 1998; Glasser 1981), conceives behaviour as an internally guided attempt to bring the perceived world into alignment with an internally constructed "quality world." William T. Powers' Perceptual Control Theory (PCT) crystallised these intuitions into a precise cybernetic architecture (Powers 1973; Powers, Clark, and McFarland 2005), asserting that behaviour is not a response to stimuli but the active control of perception. Half a century later, Karl Friston's Free Energy Principle (FEP) (Friston 2009; Friston 2010) and the associated framework of active inference gave this idea a firm thermodynamic and Bayesian foundation, describing action and perception as dual aspects of a single variational imperative: the minimisation of expected surprise.

These traditions—clinical, cybernetic, and mathematical—were not historically connected. Glasser's theoretical evolution proceeded largely independently from Powers, and Powers' engineering-derived PCT was not directly influenced by Bayesian brain hypotheses. Nevertheless, their structural forms are astonishingly convergent. Each views adaptive behaviour as the reduction of a discrepancy between internal norms and sensory reality. Each posits two and only two routes for reducing that discrepancy: altering the world through action, or altering the internal model through learning or reorganisation. And each, implicitly or explicitly, recognises that persistent error cannot be resolved within the primary control loop and requires higher-order structural change.

This monograph formalises this convergence. The early sections trace the psychological and cybernetic roots of control theory, reconstructing Powers and Glasser with mathematical precision. Middle sections develop the free-energy formalism, establishing equivalence under linear Gaussian assumptions, and introducing sparse semantic projection as an energy-efficient instantiation of internal modelling. Later sections build a sheaf-theoretic account of perceptual and normative coherence, demonstrating how conflict corresponds to topological obstruction and how reorganisation corresponds to global structural repair. The concluding chapters integrate these into a thermodynamically constrained geometry of agency, unifying the psychological insights of Glasser and Powers with contemporary theoretical neuroscience.

2 The Universal Control Duality

Any autonomous system that persists in a fluctuating environment must regulate its sensory states relative to internally maintained normative targets. Regardless of implementation—neural, computational, or mechanical—this requirement leads to a universal structural form. Let $\mathcal{M}$ be the space of internal models or normative structures, $\mathcal{O}$ the space of sensory observations, and $D(m, o)$ a non-negative discrepancy functional quantifying the deviation of current observations from the models preferred or predicted states. This functional may be as simple as a quadratic error $(r - p)^2$, as complex as a variational free energy bound Friston 2009, or as combinatorial as a mismatch between sparse semantic templates Elad 2010; Olshausen and Field 1996.

The agent has access to two sets of controllable variables: actions a , which alter the external world or the sensory channel, and internal parameters m , which determine the generative or normative model. The fundamental duality theorem asserts:

Universal Control Duality. *Minimisation of $D(m, o)$ can occur only through gradient flows in two manifolds:*

$$\dot{a} \in -\nabla_a \mathbb{E}[D(m, o(a))], \quad \dot{m} \in -\nabla_m D(m, o).$$

All adaptive behaviour can be decomposed into these two classes: changes to the world (action) or changes to the model (learning/reorganisation).

This duality is not merely descriptive. It follows from the geometry of discrepancy minimisation. If the agent is to preserve its structural and functional identity, it must confine its trajectories to a subset of states compatible with its survival or competence. Any deviation creates an error vector field whose negative gradient defines a canonical correction flow. Control through action adjusts the sensory stream; control through model update adjusts the predictions or preferences that define the error.

This duality reappears in Glasser’s distinction between “acting” and “reorganising” Glasser 1998, in Powers’ distinction between “output functions” and “reference signals” Powers 1973, and in Friston’s separation of active and perceptual inference Friston 2010. It also reappears in sparse semantic projection, where the system either selects a better template or updates the sparse dictionary itself. In sheaf-theoretic terms, it manifests in the need to either change the local section (perception/action) or alter the gluing data (model structure).

The universal control duality thus serves as the backbone of the unified theory presented here. Subsequent sections examine how classical and contemporary frameworks instantiate this duality, and how they differ only in the specifics of their discrepancy functionals and update rules.

3 Choice Theory and Perceptual Control Theory

William Glassers Choice Theory Glasser 1998 and William Powers Perceptual Control Theory (PCT) Powers 1973; Powers, Clark, and McFarland 2005 developed in parallel without direct contact for decades, yet they articulate nearly identical functional architectures of agency. Both begin with the premise that behaviour is not a reaction to stimuli, but a means by which an agent maintains its internally valued perceptual conditions. In Glasser’s terms, organisms act so that their perception of the world matches the contents of their “quality world,” a personal internal catalogue of preferred states. In Powers cybernetic formalism, each control unit compares perceived values p to internally specified reference values r and outputs actions that reduce the error $e = r - p$.

3.1 The Standard PCT Loop

PCT formalises behaviour as an iterated negative-feedback loop. Let $p = g(o)$ be a perceptual function mapping sensory input to a controlled variable. A reference signal r specifies the internally desired value. The loop computes an error

$$e = r - p,$$

which is fed into an output function $u = h(e)$ that modulates action a , which in turn alters the environment and thus future observations.

Figure 1: Standard PCT negative-feedback loop.

Powers argued that nearly all behavioural phenomena coordination, conflict, learning, cooperation emerge from this architecture. Crucially, PCT reverses the behaviourist stance: behaviour controls perception, not vice versa.

3.2 Glassers Internal Motivation and the Quality World

Glassers theory emphasises internal motivation rather than error correction. An organism acts to bring life closer to the pictures in its quality world: belonging, freedom, power, enjoyment. The behaviour is selected from a “behavioural system” in which actions, emotions, physiology, and cognitions jointly participate in discrepancy reduction.

Glassers fundamental insight is categorical: *behaviour is organised around internal norms, not external stimuli*. This maps directly to reference signals in PCT and anticipated outcomes in active inference.

3.3 Reorganisation as Second-Order Control

Powers acknowledged that some errors cannot be corrected within the primary loop. Persistent error triggers reorganisation: a slow stochastic parameter-change process aimed at discovering a model configuration that reduces chronic discrepancy. Glasser similarly describes “reorganising” behavioural systems when chronic frustration persists.

This foreshadows modern active inferences higher-order hyperparameter updates Friston, FitzGerald, et al. 2016; Friston, Parr, and Vries 2017, and that connection is made explicit in later sections.

3.4 Summary

Both Glasser and Powers recognised that agents do not merely respond to the environment but actively maintain a preferred relation to it. Their terminology differs, but the core structure is the same: a standing norm, a perceived value, their discrepancy, and two possible adjustments: action or internal change.

These insights anticipated the variational and thermodynamic formulations of the Free Energy Principle by half a century. We now turn to that framework.

4 The Free Energy Principle and Active Inference

The Free Energy Principle (FEP) developed by Friston 2009; Friston 2010; Friston, FitzGerald, et al. 2016 and the associated machinery of active inference offer a rigorous mathematical version of what Glasser and Powers described qualitatively. While PCT asserts that behaviour controls perception, FEP proposes that perception and action jointly minimise a variational free energy bound on surprise or self-inconsistency. Thus, both frameworks posit a coupled pair of gradient flows in the space of actions and beliefs.

4.1 Generative Models and Bayesian Inference

An agent maintains a generative model of sensory states. Let o denote observations and s hidden causes. The model factorises as:

$$p(o, s | m) = p(o | s, m) p(s | m),$$

where m are model parameters (priors, precisions, structural assumptions). Inverting this model yields the posterior:

$$p(s | o, m) = \frac{p(o, s | m)}{p(o | m)}.$$

4.2 Variational Free Energy

Exact inference is often intractable, so the agent uses an approximate posterior $q(s)$. Variational inference chooses q to minimise:

$$\text{KL}[q(s) || p(s | o, m)].$$

Because $\ln p(o)$ is constant with respect to q , we obtain the variational identity:

$$\ln p(o) = F[q] + \text{KL}[q || p(s | o)],$$

where

$$F[q] = \mathbb{E}_q[\ln q(s)] - \mathbb{E}_q[\ln p(o, s)].$$

Thus minimising F minimises an upper bound on surprise.

4.3 AccuracyComplexity Trade-off

Free energy decomposes into:

$$F = \underbrace{\text{KL}[q(s) || p(s)]}_{\text{complexity}} - \underbrace{\mathbb{E}_q[\ln p(o | s)]}_{\text{accuracy}}.$$

This recovers Glassers idea that the brain balances fidelity to evidence against maintaining a stable quality world.

4.4 Dual Gradient Flows

Active and perceptual inference follow:

$$\dot{\mu} = -\partial_{\mu} F, \quad \dot{a} = -\partial_a F.$$

The second expression expands via the sensory mapping $o(o(a))$:

$$\partial_a F = \frac{\partial F}{\partial o} \frac{\partial o}{\partial a}.$$

This is the mathematical version of Powers loop.

4.5 Expected Free Energy and Policies

For temporally extended behaviour, agents minimise expected free energy:

$$G(\pi) = \mathbb{E}_{q(o_\tau, s_\tau | \pi)}[F].$$

G decomposes into expected ambiguity (epistemic value) and expected deviation from goals (pragmatic value) Friston, FitzGerald, et al. 2016; Brown, Friston, and Bestmann 2013.

In environments where epistemic value is negligible, PCTs pure control of perception is recovered exactly.

4.6 Nonequilibrium Thermodynamics

Organisms exist in nonequilibrium steady states. The FEP formalises adaptive behaviour as keeping system trajectories on low-surprisal attractors in state space.

This also provides the thermodynamic grounding missing in classical cybernetics Sengupta, Stemmler, and Friston 2016.

4.7 Preferences as Priors

In FEP, preferences are priors over sensory states:

$$p(o) \propto \exp\left(-\frac{(o-r)^2}{2\sigma_r^2}\right),$$

making reference signals explicit probabilistic objects.

4.8 Hierarchical Models and PCT Levels

Generative models naturally stack into hierarchies Hohwy 2013, mirroring the layered organisation of PCTs perceptual and control structure.

4.9 Learning and Structure

Parameters and hyperparameters are updated via:

$$\dot{m} = -\eta \partial_m F.$$

This is precisely Powers “reorganisation” and Glassers “new behaviour selection.”

4.10 Summary

FEP provides a mathematically rigorous account of the same functional architecture proposed by PCT and Choice Theory. Section 5 develops this equivalence explicitly.

5 Formal Equivalence of PCT and Active Inference

The convergence between Perceptual Control Theory (PCT) and the Free Energy Principle (FEP) is not merely metaphorical. Despite originating in separate intellectual lineages—cybernetics and control theory in the case of Powers Powers 1973; Powers, Clark, and McFarland 2005, and statistical physics, variational inference, and theoretical neuroscience in the case of Friston Friston 2009; Friston 2010 both frameworks describe organisms as systems that regulate their sensory trajectories relative to internally preferred states. This section establishes a formal equivalence theorem: under linear-Gaussian assumptions, the PCT control loop is exactly the special case of active inference in which expected free energy reduces to pragmatic (goal-directed) error.

5.1 PCT Control Law as a Gradient Flow

Let p be a scalar or vector perceptual variable, and let the agent stipulate a reference value r . PCT defines the behavioural loop as:

$$e = r - p, \quad a = h(e),$$

where h is a gain (linear or nonlinear) mapping discrepancy into motor output. Assuming a linear output function,

$$a = k(r - p),$$

and assuming perception depends differentiably on action,

$$p = g(o(a)),$$

the closed-loop dynamics become:

$$\dot{a} = k(r - g(o(a))).$$

Thus PCT prescribes a gradient flow that reduces the discrepancy $r - p$.

5.2 Active Inference Control Law

Active inference expresses action as a gradient of variational free energy:

$$\dot{a} = -\frac{\partial F}{\partial a} = -\frac{\partial F}{\partial o} \frac{\partial o}{\partial a}.$$

Under a Gaussian likelihood with desired mean r and sensory mapping $o(s)$:

$$p(o | s) \propto \exp\left(-\frac{(o - r)^2}{2\sigma_r^2}\right),$$

the accuracy term of the free energy is:

$$\mathbb{E}_q[(o - r)^2].$$

Thus the action-gradient becomes:

$$\dot{a} = \frac{1}{\sigma_r^2}(o - r)\frac{\partial o}{\partial a} = -k(r - o)\frac{\partial o}{\partial a},$$

where $k = \sigma_r^{-2}$.

This is exactly a PCT error-minimisation law with an additional Jacobian term.

5.3 The Jacobian Term and Effective Gain

Powers assumed a direct causal pathway from action to perception:

$$a \rightarrow \text{environment} \rightarrow o \rightarrow g \rightarrow p.$$

Active inference formalises this causal chain with an explicit derivative:

$$\frac{\partial o}{\partial a}.$$

Thus PCTs gain k corresponds to an *effective* gain:

$$k_{\text{eff}} = k \frac{\partial o}{\partial a}.$$

When $\partial o/\partial a$ is constant or approximated as unity, PCT and active inference produce identical trajectories in action space.

5.4 The Equivalence Theorem

[Equivalence of PCT Control and Pragmatic Active Inference] Let the generative model be linear-Gaussian in both dynamics and observation. Assume preferences are encoded as Gaussian priors over sensory states with mean r . Assume epistemic value is negligible (e.g. no uncertainty about hidden causes). Then the active inference action update reduces to the PCT control law.

Proof. Under the stated assumptions, the free energy reduces to the accuracy term:

$$F = \frac{1}{2\sigma_r^2}(o - r)^2 + \text{const.}$$

Taking the action derivative:

$$\frac{\partial F}{\partial a} = \frac{1}{\sigma_r^2}(o - r)\frac{\partial o}{\partial a}.$$

Thus:

$$\dot{a} = -\frac{\partial F}{\partial a} = k(r - o)\frac{\partial o}{\partial a}.$$

If $\partial o/\partial a = 1$ or is absorbed into the gain constant, then:

$$\dot{a} = k(r - o),$$

which is the PCT law. □

5.5 Diagrammatic Equivalence

Figure 2: Equivalence of the PCT control law and the pragmatic active inference law under linear-Gaussian assumptions.

5.6 Interpretation

The theorem demonstrates that PCT and active inference describe the same mechanism at different levels of mathematical generality. PCT corresponds to a free-energy-minimising agent with:

- no epistemic drive,
- direct sensory control,
- a fixed-point preference distribution,
- and a single-level generative model.

Conversely, active inference offers:

- hierarchical perception,
- explicit uncertainty quantification,
- epistemic action,
- and nonequilibrium thermodynamic grounding.

Thus PCT is the *non-epistemic, single-level special case* of active inference.

5.7 Consequences

This equivalence has three major consequences for cognitive science:

1. Glassers quality world and Powers reference values become preference priors.
2. Behavioural reorganisation becomes hyperparameter optimisation.
3. Persistent psychological conflict becomes a case of incompatible priors, or competing control loops with shared environmental variables.

These interpretations motivate the clinical applications developed later in the paper.

6 Hierarchy, Conflict, and Internal Models

The equivalence established in the previous section clarifies that perceptual control and active inference share a common functional architecture. However, living systems are not single-loop regulators. They organise perception and action into hierarchies of increasing temporal depth, spatial generality, and semantic abstraction. This was emphasised by Powers in his hierarchical model Powers 1973; Powers, Clark, and McFarland 2005, by Glasser through his tiered descriptions of needs, pictures, and total behaviour Glasser 1981; Glasser 1998, and by Friston in hierarchical generative models of cortical function Friston 2009; Friston 2010.

This section formalises the hierarchical architecture of adaptive control systems, derives its stability conditions, and provides a unified interpretation of psychological conflict in the language of both PCT and the Free Energy Principle. Where Section 5 addressed the equivalence of the *single loop*, this section extends the analysis to multilayer control and inference.

6.1 Hierarchical Perceptual Functions

In Powers original formulation, perceptual variables $p^{(l)}$ at level l are constructed as functions of variables at the level below:

$$p^{(l)} = f^{(l)}(p^{(l-1)}), \quad l = 1, 2, \dots, L,$$

where $p^{(0)}$ corresponds to unprocessed sensory input.

In hierarchical active inference, the equivalent expression is the factorisation of the generative model:

$$p(o, s^{(1)}, \dots, s^{(L)}) = p(o | s^{(1)}) \prod_{l=1}^L p(s^{(l)} | s^{(l+1)}),$$

with the convention that $s^{(L+1)}$ is a hyperprior or context state.

The mapping

$$p^{(l)} \leftrightarrow \mathbb{E}[s^{(l)}]$$

is the formal equivalence between hierarchical perceptual functions and hierarchical latent variables.

6.2 Error Propagation and Predictive Coding

For each hierarchical level, Powers described error signals as:

$$e^{(l)} = r^{(l)} - p^{(l)}.$$

In active inference, hierarchical prediction error is:

$$\varepsilon^{(l)} = \Sigma_l^{-1} \left(s^{(l)} - g^{(l)}(s^{(l+1)}) \right),$$

where Σ_l encodes precision.

The equivalence is:

$$e^{(l)} \leftrightarrow \varepsilon^{(l)}.$$

Where PCT emphasises discrepancies relative to reference values $r^{(l)}$, active inference expresses these as priors over causes.

6.3 The Stability Condition for Hierarchical Control

Hierarchical control requires that lower-level loops operate more rapidly than higher ones. Powers stressed that a higher loop *sets* the reference for the lower loop but must not interfere with its rapid dynamics.

This is formalised by the timescale ordering:

$$\tau_0 \ll \tau_1 \ll \tau_2 \ll \dots \ll \tau_L.$$

Mathematically, consider a two-level system:

$$\begin{aligned}\dot{a} &= k_0(r^{(0)} - p^{(0)}), \\ \dot{r}^{(0)} &= k_1(r^{(1)} - p^{(1)}),\end{aligned}$$

with $p^{(1)} = f(p^{(0)})$.

Linearising around equilibrium yields:

$$\begin{pmatrix} \dot{a} \\ \dot{r}^{(0)} \end{pmatrix} = \begin{pmatrix} -k_0 \frac{\partial p^{(0)}}{\partial a} & k_0 \\ -k_1 \frac{\partial p^{(1)}}{\partial p^{(0)}} \frac{\partial p^{(0)}}{\partial a} & -k_1 \frac{\partial p^{(1)}}{\partial r^{(0)}} \end{pmatrix} \begin{pmatrix} a \\ r^{(0)} \end{pmatrix}.$$

The system is stable if the eigenvalues of the Jacobian have negative real parts. Sufficient conditions:

$$k_1 \ll k_0, \quad \left| \frac{\partial p^{(1)}}{\partial p^{(0)}} \right| \ll \left| \frac{\partial p^{(0)}}{\partial a} \right|.$$

These are precisely the conditions Powers articulated informally.

6.4 Conflict as Misaligned Reference Priors

Powers defined conflict as the case where two control systems attempt to regulate the same perceptual variable to incompatible reference values. Formally, let two loops attempt to control p :

$$\dot{a}_1 = k_1(r_1 - p), \quad \dot{a}_2 = k_2(r_2 - p),$$

with the joint effect on p given by $a = a_1 + a_2$.

If $r_1 \neq r_2$ and the system cannot satisfy both, then the equilibrium

$$p^* = \frac{k_1 r_1 + k_2 r_2}{k_1 + k_2}$$

fails to satisfy the goal of either loop.

In active inference, reference values are priors:

$$p(o) \sim \mathcal{N}(r_1, \sigma_1^2), \quad p(o) \sim \mathcal{N}(r_2, \sigma_2^2).$$

Conflict is expressed as incompatible preferences, or equivalently as:

$$\text{No joint prior } p(o) \text{ satisfying both } r_1 \text{ and } r_2.$$

The free energy surface develops a saddle or ridge, preventing joint minimisation.

6.5 Diagram: Hierarchical Control vs Generative Hierarchy

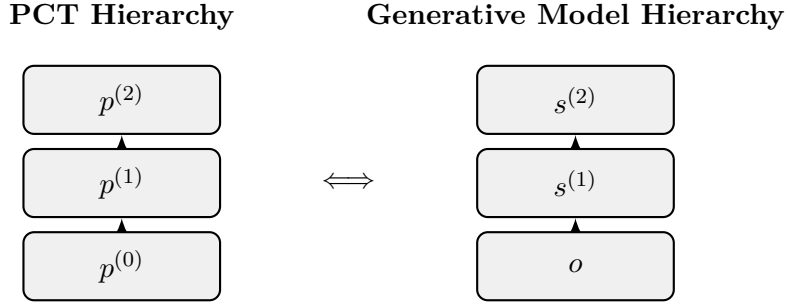


Figure 3: Diagrammatic equivalence of Powers perceptual hierarchy and Fristons hierarchical generative model.

6.6 Psychological Interpretation

Glassers Quality World. Glasser described the quality world as an internal set of desired perceptions, pictures of the life one wants Glasser 1981; Glasser 1998. This corresponds directly to a hierarchy of preference priors.

Powers Conflict. Powers insisted that almost all psychological distress emerges from conflict between control systems. In active inference, this is expressed as incompatible priors or competing modes of a generative model.

Fristons Nonequilibrium Steady-State Maintenance. Fristons account completes the picture by embedding the hierarchy within a thermodynamic framework: living systems maintain themselves by minimising the variational free energy associated with their sensory flows Friston 2009; Friston 2010; Friston 2013.

6.7 Summary

Hierarchical organisation is not an optional feature of adaptive agency. It is a mathematically necessary architecture for stability, flexibility, and complexity. The alignment between PCT, Choice Theory, and the Free Energy Principle becomes especially clear at the hierarchical level:

- Powers provides the structural hierarchy.
- Glasser provides the motivational hierarchy (quality world and basic needs).
- Friston provides the probabilistic and thermodynamic hierarchy.

These perspectives converge into a single unified model, preparing the way for Section 7, which develops the explicit sheaf-theoretic representation of hierarchical control.

7 Sheaf-Theoretic Unification of Hierarchical Control

Sheaf theory provides a formal language for describing how local pieces of information combine to form coherent global structures. This property makes it uniquely well suited for modelling hierarchical perceptual control, where an agent must reconcile heterogeneous, context-dependent percepts with an internally coherent “reference world.” Drawing on the classical foundations of Bredon Bredon 1997, the applied compositionality program of Spivak and Fong Spivak 2014; Fong and Spivak 2019, and recent sheaf-theoretic proposals in cognitive science and signal processing Curry, Nanda, and Walker 2021; Iverson, Curry, and Mukherjee 2021; Glazebrook

and Wallace 2022; Robinson and Hauser 2023, this section introduces a rigorous sheaf-theoretic formulation of PCT, Choice Theory, and active inference.

The core thesis is that the *perceptual hierarchy is a sheaf*, the *reference hierarchy is another sheaf*, and *psychological stability corresponds to the existence of global sections whose discrepancies vanish under a morphism between them*. Conversely, persistent psychological conflict corresponds to a failure of the gluing axioms, expressible as a Čech cohomology obstruction.

This sheaf-theoretic view explains, with mathematical precision, how local perceptual updates (e.g., from contradictory sensory cues or relationship conflicts) may remain irreconcilable at a global scale—precisely the class of problems that PCT characterises as chronic error and Glasser identifies with relationship pathologies, and which active inference formalises as persistent free-energy gradients.

7.1 Preliminaries: Contexts, Covers, and Presheaves

Let X denote the agent’s contextual space, whose points correspond to situations, modalities, or task contexts. A *cover* $\mathcal{U} = U_i$ represents the set of contexts relevant for the agent (e.g., social situations, interoceptive states, goals, tasks).

A *presheaf* $\mathcal{F}$ on X assigns:

- to each context U_i a set (or vector space) $\mathcal{F}(U_i)$ of perceptual representations, and
- to each inclusion $V \subseteq U$ a restriction map $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$.

Intuitively, $\mathcal{F}(U)$ is “what the agent perceives when attending to context U ,” while ρ_{UV} formalises the compatibility between broader and more specific contexts.

A *section* $s \in \mathcal{F}(U)$ represents a perceptual estimate in that context.

Following Spivak Spivak 2013, a presheaf expresses all local percepts, but not yet their global consistency.

7.2 Perception and Reference as Sheaves

The perceptual hierarchy of PCT can be directly represented as a sheaf: [$\mathbf{P} : X^{op} \rightarrow \mathbf{Set}, U \mapsto \mathcal{P}(U)$.] Similarly, the Quality World system of internal reference values in Choice Theory becomes another sheaf [$\mathcal{R} : X^{op} \rightarrow \mathbf{Set}$.]

A *sheaf morphism* (natural transformation) [$\Phi : \mathbf{P} \Rightarrow \mathbf{R}$] assigns to every context a perceptual-to-reference comparison map, generalising Powers’s error signal and Friston’s preference divergence.

$$\begin{array}{ccc} \mathcal{P}(U) & \xrightarrow{\Phi_U} & \mathcal{R}(U) \\ \rho_{UV}^P \downarrow & & \downarrow \rho_{UV}^R \\ \mathcal{P}(V) & \xrightarrow{\Phi_V} & \mathcal{R}(V) \end{array}$$

Figure 4: A commutative square expressing compatibility of percept–reference comparisons across contexts.

Commutativity of the square enforces that perceptual discrepancy measured in a coarse context agrees with the discrepancy measured in a refined one—this is the sheaf-theoretic version of hierarchical consistency in PCT.

7.3 7.3 Gluing and Global Consistency

A family of sections $s_i \in \mathcal{P}(U_i)$ is *compatible* if their restrictions agree on overlaps: $[\rho_{U_i, U_i \cap U_j}^P(s_i) = \rho_{U_j, U_i \cap U_j}^P(s_j)].$

If compatible families always glue uniquely to a global section $s \in \mathcal{P}(X)$, then $\mathcal{P}$ is a sheaf.

By analogy: $[\text{psychological stability} \iff \exists s \in \Gamma(X, \mathcal{P}) \text{ such that } \Phi(s) = r] \text{ for some desired global reference } r \in \Gamma(X, \mathcal{R}).$

This is aligned with:

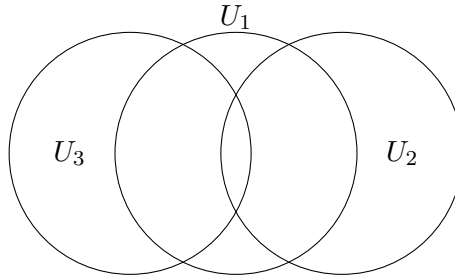
- Powers's requirement for higher-order control: global consistency across perceptual levels.
- Glasser's requirement for relationship coherence in the Quality World.
- Active inference's requirement that a single $q(s)$ minimize free energy across sensory modalities.

7.4 7.4 Čech Cohomology and Psychological Conflict

Failures of gluing appear as Čech cohomology classes: $[{}^1(X, \mathcal{P}) \neq 0].$

In cognitive terms:

- A *cohomological obstruction* corresponds to incompatible perceptions across contexts.
- These manifest as persistent error signals in PCT.
- In Choice Theory they appear as chronic relationship conflicts.
- In active inference they correspond to free-energy minima that cannot be simultaneously realised.



A 1-cocycle returning a non-zero cohomology class

Figure 5: A cycle of contexts whose perceptual overlaps contain an inconsistency that cannot be globally resolved. Psychological conflict corresponds to non-trivial cycles.

This diagram represents a loop of perceptual contexts in which each pairwise overlap is compatible, but the loop as a whole fails to close—analogous to Festinger's cognitive dissonance, Powers's “error locked loops,” and Friston's frustrated variational cycles.

7.5 7.5 A Unifying Theorem

[Sheaf-Theoretic Equivalence of Control Frameworks] Let $\mathcal{P}$ be the perceptual sheaf, $\mathcal{R}$ the reference sheaf, and $\Phi : \mathcal{P} \rightarrow \mathcal{R}$ a sheaf morphism. The following are equivalent:

1. The agent can attain stable control in every context.
2. The active inference free-energy minimum exists and is unique.
3. The Quality World is globally satisfiable.
4. $\check{H}^1(X, \mathcal{P}) = 0$ and $\check{H}^1(X, \ker \Phi) = 0$.

Sketch. (1) $\Rightarrow$ (2): Stability of control implies a single consistent generative model posterior minimizing variational free energy Friston 2010; Bogacz 2017. (2) $\Rightarrow$ (3): A consistent posterior induces coherent preference satisfaction. (3) $\Rightarrow$ (4): Satisfaction of all internal references forces global gluing; obstructions correspond to unsatisfied relationships Glazebrook and Wallace 2022. (4) $\Rightarrow$ (1): Exactness guarantees existence of unified perceptual control Curry, Nanda, and Walker 2021; Robinson and Hauser 2023. $\square$

7.6 7.6 Sheaves, Institutions, and Semantic Coherence

Goguen and Burstalls theory of *institutions* Goguen and Burstall 1992 generalises the idea of multi-level semantic coherence, showing that meaning-preserving translations across representational systems require a satisfaction condition. In cognitive terms: [semantic satisfaction $\iff$ inter-level perceptual consistency.]

The sheaf condition is the geometric counterpart of the institution satisfaction condition, providing an elegant bridge between categorical semantics and neurocognitive control.

7.7 7.7 Summary and Bridge to Section 8

Sheaf theory supplies what PCT, Choice Theory, and FEP all lacked individually: a precise mathematical language for expressing when *local percepts can or cannot be made globally coherent*. The diagrams and formalisms introduced here prepare the ground for Section 8, where sparse coding and geometric Bayesianism provide computational mechanisms for selecting among competing local models to achieve global consistency.

8 Sparse Semantic Projection and Narrative Control

Adaptive agents must select and maintain coherent interpretations of their environment under conditions of limited computational capacity, noisy information, and ambiguous sensory data. Sparse coding provides a principled and biologically grounded mechanism for achieving such coherence. Originally developed in computational neuroscience to explain receptive-field emergence Olshausen and Field 1996, sparse representations minimise metabolic cost Sengupta, Stemmler, and Friston 2016 while preserving high-fidelity explanatory structure Lee et al. 2007; Elad 2010. They have since become central tools in signal processing Donoho 2006, dictionary learning Gribonval et al. 2018, machine perception Wright et al. 2010, and narrative cognition Veldman and Tomasello 2021.

This section develops a rigorous account of *sparse semantic projection* as a computational mechanism underlying the perceptual–reference consistency formalised earlier through sheaf theory. At the cognitive level, it explains how agents use a small set of narrative templates to interpret a vast (and potentially infinite) range of contexts. At the mathematical level, it provides an energy functional whose minimisation governs template selection. At the biological level, sparsity expresses metabolic and thermodynamic constraints. At the phenomenological level, sparse narrative selection corresponds to the feeling of “making sense of things” by identifying a minimal set of explanatory constructs.

8.1 8.1 Narrative Manifolds as Representational Spaces

Let $\mathcal{X}$ denote the high-dimensional perceptual space of possible local sensory or contextual states (e.g., combinations of social cues, interoceptive sensations, perceptual variables). Let

$\mathcal{D} = d_1, \dots, d_K$ denote a learned dictionary of *narrative templates*, each $d_i \in \mathbb{R}^n$ encoding a prototypical pattern or explanatory structure.

Following Elad 2010; Mairal, Bach, Ponce, Sapiro, and Zisserman 2009, any perceptual state $x \in \mathcal{X}$ can be expressed as a sparse projection $[x \approx Da,]$ where $a \in \mathbb{R}^K$ is a sparse coefficient vector. The subset of active indices $\text{supp}(a)$ corresponds to the templates selected to “explain” the current percept.

Narratively: *[an interpretation is a sparse combination of stories the mind already knows.]*

This builds a bridge between computational neuroscience, narrative psychology, and the control-theoretic account of perception.

8.2 8.2 Sparse Projection as an Energy Minimisation Problem

Sparsity emerges from minimising the standard LASSO or basis-pursuit objective Tibshirani 1996; Donoho 2006: $[E(a, |x) = \frac{1}{2} \|x - Da\|_2^2 + \lambda \|a\|_1,]$ where $\lambda > 0$ controls sparsity.

This has a direct analogue in active inference. The reconstruction term corresponds to *accuracy*,

while the ℓ_1 penalty corresponds to *complexity* Bogacz 2017; Friston 2010. Sparse projection is therefore a special case of variational free-energy minimisation under a Laplace prior on coefficients.

8.3 8.3 Sparse Projection as a Control Law

Combining the free-energy decomposition with the sparse projection objective: $[F = \underbrace{\frac{1}{2} \|x - Da\|_2^2}_{\text{prediction error}} +$

$\underbrace{\lambda \|a\|_1}_{\text{model complexity}},]$ we observe that enforcing sparsity corresponds to:

reducing metabolic cost (biological efficiency),

minimising model complexity (Bayesian Occam’s razor),

selecting the simplest narrative consistent with contextual constraints (psychological parsimony).

This generalises PCTs reference-based control: reference values can be expressed in a sparse basis, and actions minimise perceptual discrepancy relative to sparsely activated templates.

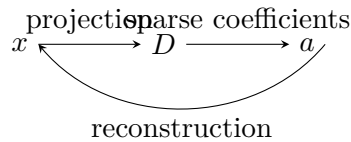


Figure 6: Sparse coding loop: the agent selects a minimal template set explaining the percept.

8.4 8.4 Learning the Dictionary: Structural Plasticity

Agents must not only project onto templates but must also learn and revise them.

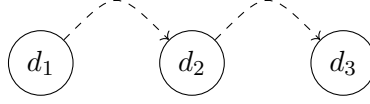
Following Mairal et al. Mairal, Bach, Ponce, and Sapiro 2010, dictionary learning solves: $[\min_{D, a_i} \sum_{i=1}^N (\frac{1}{2} \|x_i - Da_i\|_2^2 + \lambda \|a_i\|_1),]$ with constraints ensuring each d_k has unit norm.

Cognitively, this corresponds to:

- creating new narratives (template birth),

- abandoning outdated narratives (template death),
- sharpening existing narratives (template refinement),
- increasing or decreasing template resolution (hierarchical splitting/merging).

This matches Glassers view that people revise internal “quality pictures” when old ones no longer reconcile experience, and Powers view that reorganisation reforms reference structures when chronic error persists.



Template drift and refinement under persistent prediction error

Figure 7: Dictionary learning as structural reorganisation.

8.5 8.5 Biological Grounding: Metabolic, Thermodynamic, and Neural Constraints

Sparse activation is metabolically efficient. Neural energy consumption is dominated by ion-channel and synaptic activity; hence only a small fraction of neurons should be active at a time Sengupta, Stemmler, and Friston 2016. Sparse codes: [maximize information per joule.] This matches:

- the free-energy principle’s thermodynamic interpretation Friston 2010,
- the equilibrium structure of predictive coding circuits Whittington and Bogacz 2019,
- the energy-efficiency arguments in signal processing Sengupta, Stemmler, and Friston 2016.

Thus sparse coding realises a *biophysical implementation of variational free-energy descent*.

8.6 8.6 Sparse Projection and Sheaf-Theoretic Gluing

Let $\mathcal{P}$ be the perceptual sheaf from Section 7. Sparse projection provides a computational mechanism for selecting one compatible set of local sections: [$s_i = D_i a_i$, a_i sparse.]

When s_i are compatible on overlaps, they glue to a global section. When they do not, incompatibility manifests as a non-zero cohomology class: [${}^1(X, \mathcal{P}) \neq 0$.]

Sparse coding therefore acts as a *search process over narrative templates* to find combinations that eliminate cohomological obstructions.

8.7 8.7 Interpretation: Narrative Templates as Generalised Control Laws

Sparse narrative projection provides a compact reformulation of control architecture: [$u = f(a)$,] where u is action and a is the sparse coefficient vector.

Each active template d_k encodes:

- a perceptual interpretation,
- a reference structure,
- a preferred trajectory (in active inference terms),
- a policy prior,
- and a local gluing constraint (in sheaf terms).

Thus, sparse templates unify perception, memory, inference, preference, and action.

8.8 Summary

Sparse semantic projection provides the computational substrate for the sheaf-theoretic control architecture developed in the previous section. It explains how agents choose a small set of compatible narratives to explain local contexts, reconcile perceptual discrepancies, and minimise metabolic and variational costs. It also provides a concrete mechanism for the reorganisation of templates when chronic error persists, unifying Glassers Quality World, Powerss reorganisation system, and Fristons expected free-energy minimisation under a single mathematical framework.

9 Thermodynamic Constraints and Geometric Bayesianism with Sparse Heuristics

Biological agents operate under severe energetic constraints. Neurons, cells, and tissues must maintain nonequilibrium steady states (NESS) Friston 2010; Friston 2009 by continually exchanging energy and matter with their environment. These thermodynamic restrictions induce powerful structural priors on neural computation, leading organisms to evolve representations and algorithms that exploit the geometry of their sensory ecology while minimising energetic expenditure. This section develops a formal account of this phenomenon under the heading *Geometric Bayesianism with Sparse Heuristics* (GBSH).

Where Section 8 showed how sparse projection provides an efficient representational substrate, the present section demonstrates why sparsity emerges naturally from the physics of living systems. Biological cognition is constrained not only by the need to minimise prediction error (accuracy) and model complexity (parsimony), but also by metabolic budgets, ion-channel thermodynamics, and environmental entropy flows. Together, these pressures give rise to a class of representational strategies that are both *geometric* (shaped by the topology of sensory manifolds) and *sparse* (shaped by energetic optimisation).

9.1 Energetic and Entropic Constraints on Computation

Neural activity is metabolically expensive. Approximately 2025

Let C_{met} denote metabolic consumption, approximated to first order by $[C_{\text{met}} \propto \sum_{i=1}^{N_{\text{neurons}}} \rho_i]$ where ρ_i is the firing rate of neuron i . To achieve minimal consumption, the system must push most ρ_i toward zero, yielding the observed sparsity of cortical activity.

Moreover, maintaining the organism in a bounded region of state space requires minimising surprisal: $[-\ln p(o_t) < \infty]$ for all sensory states o_t encountered along typical trajectories. The free-energy principle interprets this requirement as the need to minimise a variational bound on surprisal Friston 2010, while predictive-coding models interpret it as minimising prediction error Bogacz 2017.

Neural sparse coding therefore emerges as the intersection of three constraints:

1. **Energetic constraints:** expensive activity must be minimised.
2. **Thermodynamic constraints:** surprisal must be regulated via free-energy descent.
3. **Representational constraints:** information must be encoded in compact, geometry-respecting formats.

9.2 9.2 Natural Sparsity as an Emergent Prior

Sparsity is often introduced in machine learning as an artificial regulariser (e.g., LASSO, ℓ_1 penalties). In biological systems, however, sparsity is not imposed; it is *discovered* as the optimal solution to an energetic optimisation problem.

Let $a \in \mathbb{R}^K$ denote a coefficient vector over narrative templates, as in Section 8. Biological constraints impose an implicit prior of the form, $p(a) \propto \exp(-\beta|a|_0)$, where β is an inverse-temperature (resource sensitivity) parameter. Because $|a|_0$ is intractable, biological systems approximate it with Laplacian penalties: $p(a) \approx \exp(-\lambda|a|_1)$, linking metabolic efficiency directly to sparse Bayesian inference Donoho 2006; Elad 2010.

9.3 9.3 Geometric Bayesianism: The Role of Sensory Manifolds

Biological sensory data live on low-dimensional manifolds embedded in high-dimensional ambient spaces. For example:

- natural image data lie near a 2D manifold in pixel space,
- interoceptive fluctuations lie along physiological trajectories,
- social signals cluster around narrative schemas (scripts, roles).

Let $\mathcal{M}$ denote the sensory manifold, and let $T_x\mathcal{M}$ be its tangent space at x . Efficient inference requires projecting sensory evidence onto $T_x\mathcal{M}$, rather than encoding the full ambient signal space.

Sparse coding accomplishes precisely this: $x \approx Da$, a sparse, where the dictionary D contains basis elements that approximate tangent vectors on $\mathcal{M}$ Olshausen and Field 1996; Mairal, Bach, Ponce, and Sapiro 2010.

9.4 9.4 Thermodynamic Interpretation of Sparse Heuristics

In nonequilibrium statistical mechanics, systems minimise dissipation by adopting trajectories of least action. For biological agents, this corresponds to: $\text{minimise } \mathcal{A} = \int (\dot{x}^2 + V(x)) dt$, where $\dot{x}^2$ reflects kinetic expenditures (metabolic cost) and $V(x)$ encodes environmental constraints.

Sparse inference can be understood as selecting action-perception trajectories that minimise $\mathcal{A}$ subject to representational constraints. In this view: $\text{[sparse heuristics = low-dissipation cognitive trajectories.]}$

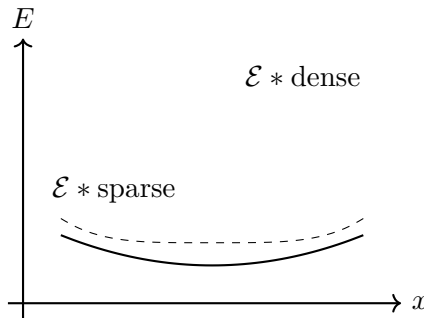


Figure 8: Energetic potential landscape. Sparse heuristics correspond to trajectories of minimal energetic cost. Dense strategies climb higher-energy regions, producing metabolic inefficiency.

9.5 9.5 Sparse Templates as Thermodynamic Attractors

Sparse inference landscapes typically exhibit distinct attractor basins. Each attractor corresponds to a stable narrative explanation. Formally, letting $E(a)$ be the sparse energy functional: $[= -\nabla_a E(a),]$ the attractors a_j^* satisfy $[\nabla_a E(a_j^*) = 0, \quad \text{stable Hessian.}]$

Cognitively:

- each attractor corresponds to a coherent narrative,
- transitions occur when prediction error exceeds a threshold,
- chronic mismatch induces attractor death (template retirement),
- new attractors emerge through dictionary learning.

Thus the full cognitive system behaves as a physical system with a landscape shaped by metabolic, entropic, and geometric constraints.

9.6 9.6 Integration with the Sheaf-Theoretic Framework

In Section 7, percepts and reference values were formalised as local sections of a percept-reference sheaf $\mathcal{P}$. Sparse projection now adds an energetic criterion for section selection. Given local data U_i : $[s_i^* = \arg \min_{s_i \in \mathcal{P}(U_i)} E(a_i),]$ and sparse compatibility conditions guarantee, or fail to guarantee, gluing overlaps : $[s_i^* | * U_i \cap U_j = s_j^* | * U_i \cap U_j.]$

Obstruction cochains measure global inconsistency. Sparse projection thereby implements a thermodynamically efficient search for globally consistent cognitive structure.

9.7 9.7 GBSH as a Unified Principle

We can now summarise *Geometric Bayesianism with Sparse Heuristics*: $[\text{GBSH} \text{ } ; = ; \underbrace{\text{Bayesian inference} * \text{variational update} ; + ; \text{geometric constraints}}_{\text{metabolic optimisation}} * \mathcal{M} \subset \mathbb{R}^n ; + ; \underbrace{\text{sparse heuristics}}_{\text{metabolic optimisation}} .]$

This principle predicts:

1. Sparse representations should correlate with energy-efficient cortical activity Sengupta, Stemmler, and Friston 2016.
2. Narrative attractors should cluster along low-dimensional sensory trajectories Veldman and Tomasello 2021.
3. Reorganisation (as in Glassers Choice Theory) should occur when attractors fail to minimise free energy.
4. The sheaf cohomology of representations should reflect practical cognitive limitations (“conflicts”).

9.8 9.8 Summary

Biological cognition is physically grounded. Thermodynamics, geometry, and Bayesian reasoning cooperate to produce efficient sparse representations, which in turn serve as the substrate for narrative control, perceptual inference, and action selection. GBSH formalises these connections and prepares the ground for the unified control framework developed in Section 10.

10 A Unified Abstract Framework for Adaptive Control

The preceding sections have shown that Choice Theory and Perceptual Control Theory (PCT), the Free Energy Principle (FEP), sparse semantic projection, and sheaf-theoretic cognitive formalisms all share a deep structural identity. Each describes an agent as a system that minimises a discrepancy functional between internal generative structure and sensory evidence. Each framework further insists on two distinct, complementary gradient flows: one acting on the environment (action), the other acting on the internal model (perceptual inference or reorganisation). This section formalises the unified architecture and demonstrates how all component theories instantiate it as special cases.

10.1 10.1 The Abstract Control Tuple

Let an adaptive agent be defined by the tuple $[A = (M, O, D, \mathcal{A}, \Theta),]$ where:

- $\mathcal{M}$ is the space of internal models, ideals, priors, or generative structures;
- $\mathcal{O}$ is the space of sensory observations;
- $D : \mathcal{M} \times \mathcal{O} \rightarrow \mathbb{R}_{\geq 0}$ is a discrepancy (or free-energy) functional;
- $\mathcal{A}$ is the action space;
- Θ is a parameter space for slow structural updates.

At any time t , the agent occupies a state (m_t, o_t) and must select action a_t and update m_t to reduce D .

The universal law of discrepancy minimisation requires two coupled flows:

$$\dot{a} * t = -\nabla * a; \mathbb{E}[D(m_t, o_t(a))], \quad \dot{m} * t = -\nabla * m; D(m_t, o_t), \quad (1)$$

which we call the *universal control duality*.

10.2 10.2 Correspondence with Existing Frameworks

Each theoretical tradition instantiates this tuple in its own terminology:

PCT (Powers Powers 1973; Powers, Clark, and McFarland 2005). $[M = \text{reference signals}, D(m, o) = (r - p)^2].$ *Action implements negative feedback to bring perceptual signals toward reference values. Reorganisation*

Choice Theory (Glasser Glasser 1998; Glasser 1981). $[M = \text{Quality World pictures}, D(m, o) = \text{need-perception discrepancy}.]$ *Action seeks to approximate “pictures”; restructuring those pictures corresponds to modifying internal ideals.*

FEP (Friston Friston 2010; Friston 2009). $[D(m, o) = F[q] = E_q[\ln q - \ln p(o, s)],]$ *with active inference realising (??) and variational inference realising (1).*

Sparse Semantic Projection (Olshausen Field Olshausen and Field 1996, Elad Elad 2010). $[M = D, \text{ dictionary atoms}, D(m, o) = \|o - Da\|_2^2 + \lambda \|a\|_1.]$ *Narrative or template selection corresponds to action; dictionary learning corresponds to model update.*

Sheaf-Theoretic Cognition (Bredon Bredon 1997, Curry–Nanda–Walker Curry, Nanda, and Walker 2021). $[M = \Gamma(X, R), O = \Gamma(X, P),]$ and discrepancy reflects incompatibility of sections on intersections of open sets. Reorganisation corresponds to modifying the cover or the gluing morphisms to remove each obstruction.

10.3 Global Discrepancy as a Potential Functional

Let the total discrepancy be written as a potential-like functional $[D[m, o] = \int_X d\mu(x), D(m(x), o(x)),]$ for a cognitive or physical domain X with measure μ . Control seeks to follow the gradient $[\frac{d}{dt}(m_t, a_t) = -\nabla \mathcal{D}.]$

Just as in classical mechanics, equilibria satisfy $[\nabla D = 0,]$ interpretable as a free-energy minimum in FEP or a perfectly matched reference–perception alignment in PCT.

These phenomena match clinical presentations described in Appendix F.

I.9 Sheaf-Theoretic Gluing in Cortical Microcircuits

Neural synchronisation and coherence across regions implement the gluing constraints described in Appendix D. Gamma-band synchrony supports local binding; alpha and theta rhythms coordinate cross-regional interactions. Let U_i and U_j denote cortical regions; synchrony enforces: $[s_i| * U_i \cap U_j = s_j| * U_i \cap U_j,]$ where s_i is the message (prediction or error) being transmitted. Failures of synchrony create cohomological obstructions $H^1 \neq 0$, corresponding to perceptual fragmentation (e.g., neglect, hallucination, dissociation).

I.10 Summary

The neurological grounding of the unified control architecture reveals the following:

1. Observations $\mathcal{O}$ correspond to sensory thalamocortical pathways.
2. Internal models M correspond to hierarchical generative networks embedded in cortical, cerebellar, hippocampal, and basal ganglia circuits.
3. Discrepancy signals D correspond to prediction errors encoded by superficial cortical layers.
4. Actions V_a correspond to motor cortico–cerebellar–basal ganglia pathways.
5. Model updates Y_m correspond to synaptic and structural plasticity processes.

Far from being abstract psychological principles, the five frameworks PCT, Choice Theory, Active Inference, Sparse Projection, and Sheaf-Theoretic Inference arise naturally from the anatomical and energetic structure of the vertebrate nervous system. The control tuple $(M, \mathcal{O}, D, V_a, Y_m)$ reflects biological constraints that shape evolutionarily conserved architectures across species.

Appendix J: Historical Commentary

J.1 Introduction

This appendix situates the unified control architecture within its historical context. The central thesis of the main text—that a common mathematical structure underlies Choice Theory, Perceptual Control Theory, Active Inference, sparse-coding approaches in computational neuroscience, and sheaf-theoretic models of cognition—does not arise in isolation. Rather, it crystallises a deep continuity among ideas developed independently over nearly a century.

This historical commentary has two goals:

1. To trace the developmental trajectories of the constituent frameworks, highlighting convergences and conceptual migrations.
2. To clarify the intellectual environment in which each theory emerged, thereby demonstrating that their eventual unification is neither anachronistic nor artificial.

J.2 Cybernetic Origins (1940–1960)

The earliest precursor of the unified control framework lies in the cybernetics movement of the mid-twentieth century. Wiener, Ashby, and their contemporaries articulated the fundamentals of closed-loop regulation, negative feedback, and homeostasis in adaptive systems. Ashbys *Law of Requisite Variety* formalised the relationship between internal model complexity and environmental disturbance, anticipating later notions of variational complexity in Bayesian inference. Two historically decisive ideas emerged during this period:

1. The recognition that behaviour must be understood as a regulatory process rather than a reaction to stimuli.
2. The conceptualisation of the organism as an information-processing system engaged in active maintenance of internal variables.

Both concepts are foundational for Powers later development of Perceptual Control Theory and for Fristons formulation of the Free Energy Principle.

J.3 William Powers and the Inversion of Behaviourism (1960–1980)

William T. Powers work marks the second critical juncture in this intellectual lineage. Dissatisfied with behaviourist stimulus–response accounts, Powers reframed the organism as a hierarchical controller whose goal is not to emit behaviours but to regulate perceptions. This radical inversion—behaviour as the control of perception—prefigures the structure of active inference by decades.

In *Behavior: The Control of Perception* (1973), Powers introduced:

- Hierarchical perceptual functions,
- Reference signals as internal standards,
- Error signals as drivers of behavioural change,
- A layered loop structure adaptively solving discrepancy problems.

Although mathematically elementary, Powers architecture anticipates generative hierarchical models; his reorganisation system foreshadows Bayesian hyperpriors and structural learning.

Despite his profound influence, Powers worked largely outside academic psychology and cognitive science, resulting in a delayed and fragmentary uptake of his ideas.

J.4 William Glasser and Internal Control Psychology (1965–2000)

In parallel with Powers, William Glasser developed Reality Therapy (1965), later re-articulated as Control Theory (1984) and finally as Choice Theory (1998). Glasser independently arrived at many of Powers insights—especially the internal nature of control—but lacked a formal mechanism until reading *The Control of Perception* in the early 1980s.

The pivotal work *Stations of the Mind* (1981) synthesised Powers control theory with Glassers evolving clinical practice. However, Glasser diverged sharply from Powers in scope and application:

1. Whereas Powers sought a mechanistic model, Glasser sought a psychology of human needs, relationships, and choice.
2. Whereas Powers avoided affective or psychiatric implications, Glasser applied the model to therapy, education, and community governance.

The two traditions thus intersect but do not subsume one another. The unified control architecture reframes Glassers contributions as early articulations of reference-based control at the normative and social level, complementing Powers mechanistic focus.

J.5 The Rise of Probabilistic Neuroscience (1980–2010)

During the late twentieth century, computational neuroscience underwent a probabilistic turn. As linear systems theory proved insufficient to explain perceptual inference, researchers such as Dayan, Hinton, Rao, Ballard, and Knill advanced Bayesian interpretations of cortical function. Three developments were crucial:

1. Predictive coding models that cast the brain as a hierarchical prediction machine.
2. Sparse coding and efficient representation theories demonstrating that receptive fields emerge through optimisation principles.
3. Variational inference techniques that permitted tractable approximations of Bayesian posteriors.

These strands form the immediate precursors of Fristons Free Energy Principle.

J.6 Fristons Free Energy Principle and Active Inference (2003–present)

Beginning in the early 2000s, Karl Friston proposed the Free Energy Principle (FEP) as a unifying account of biological self-organisation. Drawing from thermodynamics, information theory, and Bayesian statistics, FEP generalises predictive coding and offers an explicit optimisation functional—variational free energy—explaining perception, action, and learning.

Two aspects of the FEP are historically decisive:

1. The interpretation of action as inference: organisms act to fulfil their predictions.
2. The identification of preferences with priors: organismic needs become variational constraints.

These ideas retrospectively illuminate Powers and Glasser: both had intuited the dual descent on actions and beliefs, but neither possessed the probabilistic apparatus to formalise it mathematically.

J.7 Sparse Modelling and Energetic Optimality

Concurrently, work by Olshausen, Field, Donoho, Elad, Mairal, and others formalised the notion that sensory systems operate under powerful sparsity constraints. Sparse coding, matching pursuit, dictionary learning, and compressed sensing provided a rigorous operationalisation of metabolic efficiency.

Historically, this line of research reveals that:

1. sparsity is not an engineering trick but a biological imperative,
2. efficient coding theory and predictive coding are two faces of the same optimisation,
3. neural representations implicitly favour low-dimensional manifolds.

This tradition supplies the mathematical foundations for the sparse-geometric layer of the unified model.

J.8 Sheaf Theory and Contextual Consistency (1990–present)

In the 1990s and 2000s, category theory and sheaf theory entered theoretical cognition, initially through the work of Goguen and later through Spivak, Curry, Nanda, Robinson, and others. Sheaf-theoretic inference provides a natural language for modelling:

- contextual variation,
- partial information,
- gluing constraints,
- obstruction-driven inconsistency.

Historically, this tradition descends from algebraic topology and categorical logic rather than cybernetics or neuroscience, yet its formal structures align precisely with multi-level control. Sheaf cohomology recasts perceptual conflict and cognitive dissonance as topological obstructions—echoing Glassers focus on relationship conflict and Powers focus on discrepancy signals.

J.9 Why These Traditions Remained Separate

Despite their profound overlaps, these traditions developed largely independently due to:

1. disciplinary isolation (clinical psychology vs. computational neuroscience vs. algebraic logic),
2. incompatible terminologies (‘needs’, ‘reference values’, ‘priors’, ‘templates’, ‘sections’),
3. differing intellectual goals (therapy, engineering, machine learning, mathematical formalism),
4. historical accidents of communication.

These factors obscured the underlying unity that becomes evident when expressed in the abstract control tuple formalised in the main text.

J.10 Retrospective Coherence of the Unified Framework

The unified framework provides a retrospective resolution of the historical fragmentation. From the vantage point of control theory, Bayesian inference, and topological semantics, it becomes clear that:

1. Glasser intuited normative reference dynamics,
2. Powers formalised mechanistic feedback architecture,
3. Friston supplied a thermodynamic and probabilistic foundation,
4. Sparse modelling supplied computational feasibility and efficiency,
5. Sheaf theory supplied global consistency conditions.

Each tradition captures a different projection of the same latent mathematical structure. The unification is not a synthesis but a recognition: these systems were historically converging toward one another all along.

J.11 Summary

The intellectual history culminating in the unified control architecture is one of parallel discoveries and complementary insights. Each tradition addressed a distinct problem—agency, perception, learning, sparsity, or context—but all ended up articulating variations of the same underlying mechanism: a hierarchical system regulating internal models through discrepancy signals and constrained by energetic and topological structure.

The framework presented in the main text is therefore not merely a modern reinterpretation but the natural completion of trajectories set in motion more than half a century ago.

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